RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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**R V College of Engineering**

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

**IV Semester Regular/Supplementary Examinations June/July- 2025**
**Course :** CAD/CAM & Robotics-IM343AI

**Time : 3 Hours** **Maximum Marks : 100**

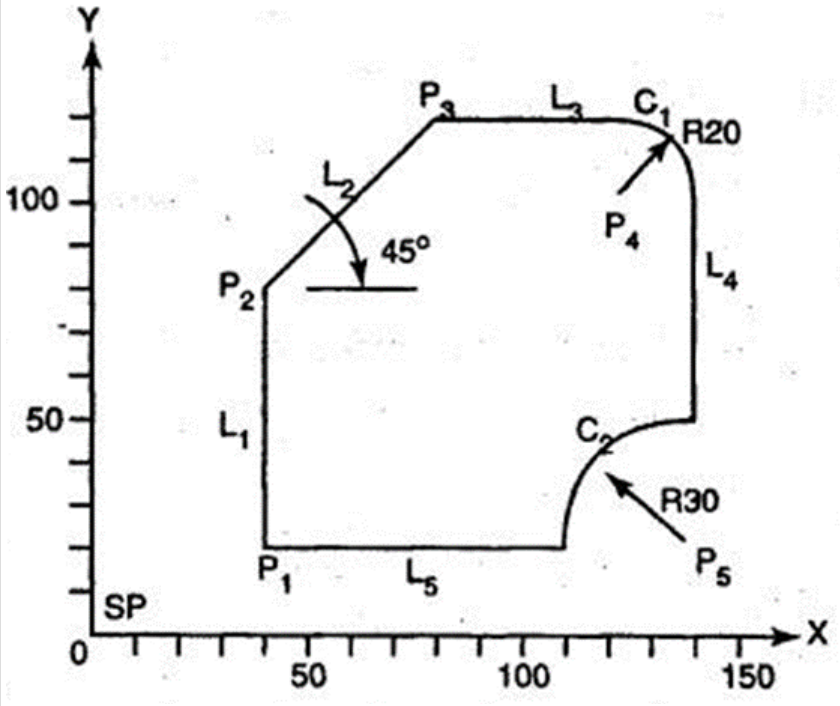
***Instructions to the students***

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

**Part A**

| Question No | Question | M | CO | BT |
|---|---|---|---|---|
| 1.1 | Why is conceptual design an important phase in the Shigley model? | 02 | 1 | 1 |
| 1.2 | What is the purpose of a line drawing algorithm? | 02 | 1 | 1 |
| 1.3 | What is a part program in NC machining? | 02 | 2 | 2 |
| 1.4 | What is tool offset in CNC machining? | 02 | 2 | 2 |
| 1.5 | What is mechanization? | 02 | 2 | 2 |
| 1.6 | List two motivating factors for robot usage in industries | 02 | 3 | 1 |
| 1.7 | The power source of a robot can be electric, hydraulic, or _____. | 01 | 3 | 1 |
| 1.8 | What is meant by payload in robotics? | 02 | 4 | 2 |
| 1.9 | Give two examples of exteroceptive sensors | 02 | 4 | 1 |
| 1.10 | What is lead-through programming? | 02 | 4 | 1 |
| 1.11 | A _____ is a device at the end of a robot arm, designed to interact with the environment | 01 | 4 | 1 |

**Part B**

| Question No | Question | M | CO | BT |
|---|---|---|---|---|
| 2a | Use Bresenham's Line Algorithm to rasterize a line from (10, 15) to (20, 25). Show all intermediate steps. | 10 | 1 | 4 |
| 2b | Discuss the various applications of CAD in different industries **s**uch as mechanical, civil, and electrical engineering. | 06 | 1 | 3 |
| 3a | Analyze the effect of incorrect tool offset values in CNC milling programs. | 06 | 2 | 1 |
| 3b | Compare APT programming with manual NC programming in terms of flexibility, accuracy, and complexity. | 10 | 2 | 3 |
| | **OR** | | | |
| 4a | Prepare a program for milling the given component by milling process using APT.<br><br>Post Processor: Machine Mill, 1<br>Feed rate: 60 mm / min<br>Spindle speed: 500 rpm<br>Cutter diameter: 5 mm<br>Tolerance: ± 0.02 mm<br>Thickness: 10 mm<br><br>  <br> | 16 | 2 | 4 |

| | | | | |
|---|---|---|---|---|
| 5a | Explain the social issues associated with automation | 08 | 2 | 1 |
| 5b | Explain reasons for adopting automation in industries. | 08 | 3 | 3 |
| | **OR** | | | |
| 6a | Explain how automation contributes to sustainable manufacturing | 08 | 3 | 4 |
| 6b | Explain the different types of automation with examples. | 08 | 2 | 2 |
| 7a | Discuss the selection criteria for industrial robots in manufacturing applications. | 08 | 3 | 2 |
| 7b | Explain the robot wrist and end-of-arm tooling (EOAT) with examples | 08 | 3 | 2 |
| | **OR** | | | |
| 8a | A family-owned metal fabrication workshop is evaluating whether to automate its manual cutting and welding operations. However, the owner is hesitant. Analyze the reasons why automation might not be suitable for such a workshop. Provide alternatives. | 10 | 4 | 4 |
| 8b | Explain the classification of robots based on configuration | 06 | 3 | 1 |
| 9a | Describe in detail the role of sensors in robotics and how they improve robot performance. | 08 | 4 | 3 |
| 9b | How do exteroceptive sensors support robotic adaptability in dynamic environments? | 08 | 4 | 2 |
| | **OR** | | | |
| 10a | A bakery needs a robot to pick and place soft pastries into boxes. Recommend an end effector and justify it based on hygiene, delicacy, and cycle time. | 10 | 4 | 5 |
| 10b | Explain on-line robot programming and its advantages | 06 | 4 | 1 |

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# RV COLLEGE OF ENGINEERING®

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**IV Semester B. E. Examinations Sept/Oct – 2024**

**Industrial Engineering and Management**

## CAD/CAM AND ROBOTICS

***Time: 03 Hours*** ***Maximum Marks: 100***

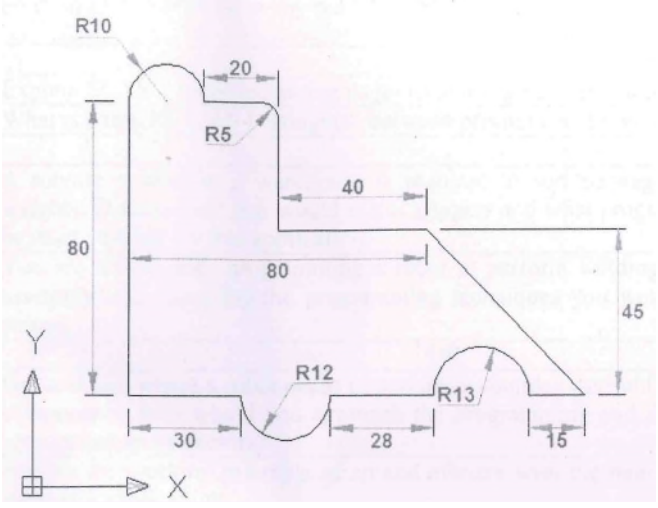
***Instructions to candidates:***

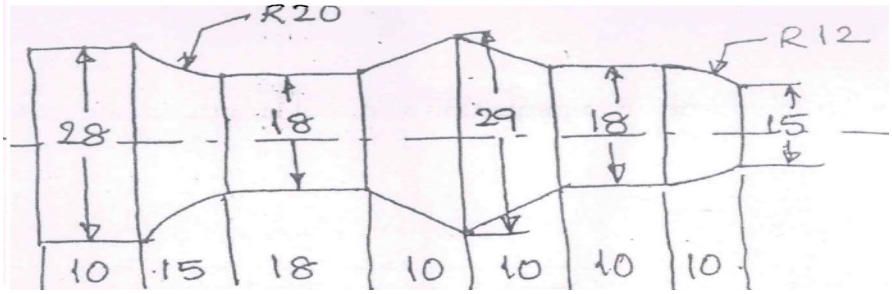
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

**PART-A**

| | | | M | BT | CO |
|---|---|---|---|---|---|
| 1 | 1.1 | What is rendering in computer graphics? | 02 | 2 | 1 |
| | 1.2 | What tis the Z-buffer algorithm used for? | 02 | 2 | 1 |
| | 1.3 | List the main elements of an NC system. | 02 | 2 | 2 |
| | 1.4 | Identify two keys components of a DNC system. | 02 | 1 | 2 |
| | 1.5 | How does mechanization differ from automation? | 02 | 1 | 2 |
| | 1.6 | What are the primary goals of automation? | 02 | 2 | 1 |
| | 1.7 | Define zeroth law of Robotics. | 02 | 1 | 1 |
| | 1.8 | List any two features of fixed automation. | 02 | 1 | 2 |
| | 1.9 | Define pitch, yaw and roll? | 02 | 2 | 2 |
| | 1.10 | What is meant by work envelop? | 02 | 1 | 1 |

**PART-B**

| | | | | | |
|---|---|---|---|---|---|
| 2 | a | Consider a scenario where you are designing a complex mechanical part using CAD. Discuss how you would utilize the different stages of the Shigley Model to ensure a successful design. | 08 | 2 | 1 |
| | b | Express a line from (5,5) to (13,9) on a raster screen using Bresenham's Line Drawing Algorithm. Show the result in a Cartesian graph. | 08 | 3 | 2 |
| | | | | | |
| 3 | a | Figure 3a below shows the final profile to be generated on a MS by using a CNC milling center. Write CNC program and show the tool path.   Fig 3a | 12 | 3 | 3 |

| | | | | | |
|---|---|---|---|---|---|
| | b | What key factors should be considered during the Problem? Definition stage of the Shigley Model when designing a mechanical part? | 04 | 2 | 2 |
| | | **OR** | | | |
| 4 | a | Explain the APT statements:<br>(i) GOTO and GODLTA<br>(ii) INTOL and OUTTOL<br>(iii) TANTO and GOBLACK | 06 | 3 | 2 |
| | b | Write a manual part program for the component shown below.<br>  <br><br>Fig. 4b | 10 | 3 | 2 |
| | | | | | |
| 5 | a | Explain the concept of Low-Cost Automation (LCA) and its significance in small and medium enterprises (SMEs). | 08 | 2 | 2 |
| | b | Imagine you are managing a small manufacturing unit. How would you implement Low-Cost Automation to improve productivity without incurring high expenses? Provide specific example. | 08 | 4 | 3 |
| | | **OR** | | | |
| 6 | a | A factory is considering the shift from manual assembly to automated assembly lines. Discuss the issues they might face during this transition and how these challenges can be mitigated. | 08 | 4 | 4 |
| | b | How would you design a strategy for automation in a factory that currently relies heavily on manual labor? Outline the key steps involved. | 08 | 3 | 3 |
| | | | | | |
| 7 | a | Briefly explain the various components of an industrial robot. State the main function of each of the components. | 08 | 2 | 3 |
| | b | What do you understand by degree of freedom (DOF)? How many DOFs are required to position an end effector at any point in 3-D space? | 08 | 2 | 2 |
| | | **OR** | | | |
| 8 | a | Explain SCARA robot indicating its joints and degree of freedom with the applications. | 08 | 2 | 3 |
| | b | What is a robotic joint? Distinguish between prismatic and revolute joints. | 08 | 2 | 3 |
| | | | | | |
| 9 | a | A robotic system in a warehouse is required to sort packages of varying sizes and weights. Discuss how you would select grippers and what programming approach would be most suitable for this application. | 08 | 3 | 3 |
| | b | You are tasked with programming a robot to perform welding tasks in an automotive assembly line. Describe the programming techniques you would use and justify your choice. | 08 | 3 | 4 |

| | | | | | |
|---|---|---|---|---|---|
| | | **OR** | | | |
| 10 | a | In a scenario where a robot needs to perform a complex assembly task involving multiple components, how would you approach the programming and integration of the robot's sensors and end effectors? | 08 | 4 | 4 |
| | b | Explain the working principle of an end effector with the help of a sketch and give its important applications. | 08 | 2 | 3 |

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# RV COLLEGE OF ENGINEERING®

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**IV Semester B. E. Examinations October-2023**

## Industrial Engineering and Management

## CAD/CAM AND ROBOTICS

***Time: 03 Hours*** ***Maximum Marks: 100***

***Instructions to candidates:***

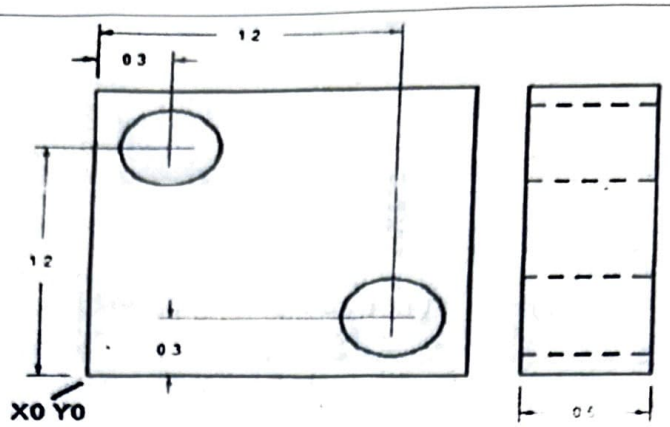
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

### PART-A

| | | | |
|---|---|---|---|
| 1 | 1.1 | Define Robot Programming. | 02 |
| | 1.2 | List any two advantages of suction grippers. | 02 |
| | 1.3 | Identify any two objectives of an Industrial Robot. | 02 |
| | 1.4 | If given an opportunity to build a robot to be used in the department of *IEM* at *RVCE*, what problem would the robot solve and what would be its functionality? | 02 |
| | 1.5 | _____ and _____ are special purpose tool which are used to facilitate production (machining, assembling and inspection operations) when work piece is based on the concept of interchangeability according to which every part will be produced within an established tolerance. | 02 |
| | 1.6 | Students who carry out productivity studies at industries as part of major project sometimes tend to conclude that one or set of robots can be used to automate the entire process and enhance productivity by multifold and cost of commissioning it can be recovered in a very short period by compensating the salaries of the employee who can be removed from the process. Review such inference from the standpoint of India as a country. | 02 |
| | 1.7 | Identify two disadvantages of automation. | 02 |
| | 1.8 | Identify two applications of robots in industrial sector and service sector. | 02 |
| | 1.9 | Tactile sensors are also called as _____ and are sensitive to _____. | 02 |
| | 1.10 | State the Zeroth Law of robotics. | 02 |

### PART-B

| | | | |
|---|---|---|---|
| 2 | a | Explain the Shigley model with a block diagram. | 10 |
| | b | Enumerate any six benefits of *CAD* in automation industry. | 06 |
| | | | |
| 3 | a | Write a *CNC* peck drilling program for the profile given in Fig 3a. The cutting tool used is slot drill cutter and the depth of cut is 5*mm*. | |

| | | | |
|---|---|---|---|
| | |  Fig 3a | 10 |
| | b | Discuss the various elements of a typical Numerical Control System. | 06 |
| | | **OR** | |
| 4 | a | Explain with the help of an example<br>i) Linear interpolation<br>ii) Circular interpolation. | 06 |
| | b | Explain any five types of Distributed Numerical Control Systems. | 10 |
| | | | |
| 5 | a | Identify major socio-economic considerations favoring automation. | 08 |
| | b | With example, state the difference between mechanization and automation. | 08 |
| | | **OR** | |
| 6 | a | Explain how low-cost automation is achieved in manufacturing industry. | 10 |
| | b | Describe six major issues for Automation in Factory Operations. | 06 |
| | | | |
| 7 | a | The robot has to do the following functions. Recommend the types of sensors it should be fitted with to carry out the operations successfully. Also mention one specification of the recommended sensor.<br>Functionality 1: A binary output signal when it comes in contact with an object.<br>Functionality 2: To sense and indicate presence of any object without physical contact.<br>Functionality 3: To identify the distance of sensor from the object.<br>Functionality 4: To estimate the speed with which a manipulator is moved. | 08 |
| | b | Explain the advantages and disadvantages of vacuum/suction grippers. | 08 |
| | | **OR** | |
| 8 | a | Explain the characteristics of an industrial robot. | 10 |
| | b | How are the robots classified? Explain. | 06 |
| | | | |
| 9 | a | Assume that the robot must pick up objects from a chute and place them in successive boxes. Choose the sequence of robot activity for which robot programming will have to be written at a later stage. | 08 |
| | b | Explain any two robot programming techniques. | 08 |
| | | **OR** | |
| 10 | a | Explain the factors considered for selection of gripper. | 10 |
| | b | Write short notes on types of sensors in robots. | 06 |

A

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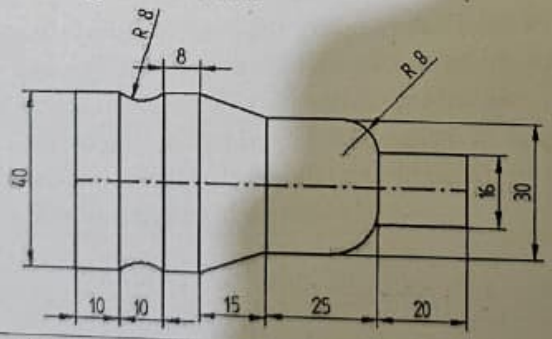
## Academic year 2024-2025 (EVEN Sem)

DEPARTMENT OF

# INDUSTRIAL ENGINEERING & MANAGEMENT

| Date | 1st April 2025 | Maximum Marks | 10 + 50 |
|---|---|---|---|
| Course Code | IM343AI | Duration | 120 Min |
| Sem | IV Semester | CIE –I | |
| CAD/CAM & ROBOTICS | | | |

**Note:**
1. Answer all the Questions.

| Sl. No. | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1.1 | List any two advantage of using CAD over manual drafting? | 02 | L2 | CO1 |
| 1.2 | How does DNC differ from CNC? State any two. | 02 | L1 | CO2 |
| 1.3 | List any two reasons for implementing CAD in engineering and product design? | 02 | L2 | CO2 |
| 1.4 | What are the key components of a DNC system? | 02 | L2 | CO1 |
| 1.5 | What are two essential elements of an NC system? | 02 | L1 | CO2 |
| | **Part –B** | | | |
| 1 | Discuss the various applications of computers in the design process with its various stages and their importance in engineering design. | 10 | L1 | CO2 |
| 2 | According to the type of machine, NC machine are classified into three types explain the classification with the help of Case examples. | 10 | L2 | CO1 |
| 3 | Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. Using G & M Codes   | 10 | L4 | CO2 |
| 4 | Develop a complete APT Part Program for machining the given part. (Profile Milling only). Tool-Slot drill Cutter of ϕ10mm, Cutting Speed-1500, Feed- 60, Depth of cut=6mm | 20 | L4 | CO1 |

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DEPARTMENT OF

# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | | |
|---|---|---|---|
| **Date** : 5th May 2025 | **CIE – II** | **Max. Marks** : 10 + 50 | |
| **Semester** : IV | **UG** | **Duration** : 30 + 90 Min | |
| **Course Title: CAD/CAM & ROBOTICS** | | **Course Code** : IM343AI | |

**Note:**

1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | What is rendering in computer graphics? | 2 | 1 | 1 |
| 2. | Why is the complexity of Bresenham's line drawing algorithm less than that of DDA line drawing algorithm? | 2 | 2 | 1 |
| 3. | List any two features of fixed automation | 2 | 1 | 1 |
| 4. | What is the difference between Mechanization and Automation? | 2 | 2 | 3 |
| 5. | List two common reasons for adopting Automation. | 2 | 2 | 3 |
| | **Part – B** | | | |
| 1 | Express a line from (5, 5) to (13, 9) on a raster screen using Bresenham's Line Drawing Algorithm. Show the result in a Cartesian graph. | 10 | 3 | 1 |
| 2 | What is scan conversion in computer graphics? Explain how lines and polygons are scan converted. | 10 | 2 | 1 |
| 3 | After automation in a packaging unit, several workers lost their jobs. There's rising concern about unemployment and skill mismatch. Discuss the social challenges caused by automation. Suggest how these issues can be mitigated. | 10 | 3 | 3 |
| 4 | A beverage bottling plant uses high-speed equipment that fills, caps, and labels bottles on a continuous production line. The system is dedicated to a single type of bottle and product. Identify and explain the type of automation used in this plant. Discuss its advantages and limitations in high-volume production. | 10 | 3 | 3 |
| 5 | A manufacturing plant wants to implement automation in stages to minimize disruption and investment risk. Outline different strategies for automation and explain which would best suit a gradual transition. | 10 | 2 | 3 |

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# INDUSTRIAL ENGINEERING & MANAGEMENT

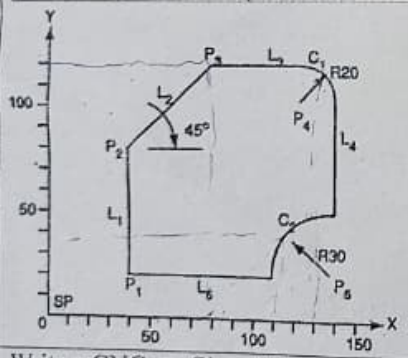
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|---|---|---|
| **Date : 25th May 2026** | **CIE - II** | **Max. Marks : 10 + 50** |
| **Semester : IV** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: CAD/CAM & ROBOTICS** | | **Course Code : IM343AI** |

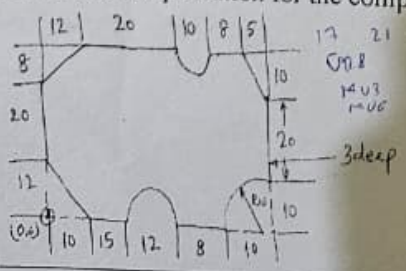
**Note:**

1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | Define point-to-point NC system. | 2 | 1 | 2 |
| 2. | What is an APT program in CNC machining? | 2 | 1 | 2 |
| 3. | Differentiate between mechanization and automation. | 2 | 2 | 3 |
| 4. | List two drawbacks of not adopting automation. | 2 | 2 | 3 |
| 5. | Name two areas where low-cost automation is preferred | 2 | 3 | 3 |
| | **Part – B** | | | |
| 1 | Discuss the different types of automation and their applications in manufacturing. | 10 | 2 | 3 |
| 2 | Discuss the various strategies adopted by industries for successful automation implementation. | 10 | 2 | 3 |
| 3 | Prepare APT program for the given component by milling process. | 15 | 3 | 2 |
| 4 | Write a CNC profile milling program for the component shown below. Take left hand tool radius compensation for the component shown in Fig | 15 | 3 | 2 |

| Post Processor | Machine Mill, 1 |
|---|---|
| Feed rate | 60 mm / min |
| Spindle speed | 500 rpm |
| Cutter diameter | 5 mm |
| Tolerance | ± 0.02 mm |
| Thickness | 10 mm |



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1RV23IM037 A

DEPARTMENT OF

# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | |
|---|---|---|
| **Date : 2nd June 2025** | **Improvement CIE** | **Max. Marks : 10 + 50** |
| **Semester : IV** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: CAD/CAM & ROBOTICS** | | **Course Code : IM343AI** |

**Note:**

1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | The physical boundary within which a robot operates is its _____ | 1 | 1 | 3 |
| 2. | The robot joint that provides rotational motion is called a _____joint | 1 | 2 | 4 |
| 3. | A _____ is a device at the end of a robot arm, designed to interact with the environment | 1 | 1 | 4 |
| 4. | Industrial robots typically follow a _____ cycle for repeated operations | 1 | 2 | 4 |
| 5. | _____ sensors gather information from the robot's external environment | 1 | 2 | 4 |
| 6. | Which type of programming is suitable for complex and repeatable tasks? | 1 | 3 | 4 |
| 7. | What is teach pendant in robotics? | 2 | 2 | 4 |
| 8. | What is a gripper in robotics? | 2 | 1 | 3 |
| | **Part – B** | | | |
| 1 | Describe the selection criteria and types of grippers used in robotics. | 10 | 2 | 3 |
| 2 | Explain the characteristics of industrial robots and how they impact manufacturing industries. | 10 | 2 | 4 |
| 3 | Explain the robot wrist and end-of-arm tooling (EOAT) with examples. | 10 | 2 | 4 |
| 4 | Describe in detail the role of sensors in robotics and how they improve robot performance. | 10 | 1 | 4 |
| 5 | Identify any four robot programming languages. Name the important requirement of programming languages. | 10 | 1 | 4 |

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DEPARTMENT OF

# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | |
|---|---|---|
| **Date : 18th June 2026** | **Improvement CIE** | **Max. Marks : 10 + 50** |
| **Semester : IV** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: CAD/CAM & ROBOTICS** | | **Course Code : IM343AI** |

**Note:**
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | State the primary purpose of an industrial robot? | 1 | 1 | 3 |
| 2. | State the First Law of Robotics. | 1 | 2 | 4 |
| 3. | What are the three basic wrist motions? | 1 | 1 | 4 |
| 4. | Mention any two applications of proprioceptive sensors? | 1 | 2 | 4 |
| 5. | Why are robot languages required? | 1 | 2 | 4 |
| | **Part – B** | | | |
| 1 | Define a robot and explain the characteristics of an industrial robot. | 10 | 2 | 3 |
| 2 | A consumer electronics company observes quality variations in manual assembly operations. The management is evaluating whether to replace certain manual assembly tasks with robotic systems. Compare the performance of human manipulators and robot manipulators in this situation and justify your recommendation. | 10 | 3 | 4 |
| 3 | A manufacturing company uses robots to handle fragile glass sheets during production. Frequent breakage is occurring because the current gripper applies excessive force.<br>a) Which type of end effector should be selected?<br>b) Explain why it is suitable.<br>c) Describe the role of the robot wrist during this operation. | 10 | 3 | 4 |
| 4 | A logistics company has deployed autonomous mobile robots to transport materials between storage racks and dispatch areas. Recently, the robots have been colliding with obstacles and occasionally taking incorrect routes, causing delays in operations.<br>Questions:<br>a) Identify the type of sensors required to avoid collisions.<br>b) Classify the sensors as internal or external sensors.<br>c) Explain the working principle of any two sensors suitable for this application.<br>d) Discuss how the selected sensors improve the efficiency and safety of warehouse operations. | 10 | 3 | 4 |
| 5 | Explain the three methods for programming a robot | 10 | 2 | 4 |